

StrataAnalytics · Enterprise Intelligence Platform

Full-stack web application & interactive data visualization platform engineered with React.js, Python, and MySQL.

Role: Full-Stack Developer | Timeline: 2022 · 2023 | Tools: React.js, Python, MySQL, REST APIs, Chart.js

01 Project Overview & Problem Statement

Enterprise operations teams were struggling with scattered spreadsheet exports and delayed batch reports. Decision-makers lacked real-time visibility into revenue trajectories, user cohort churn, and regional sales performance.

StrataAnalytics was engineered to unify distributed transactional databases into a real-time, responsive web intelligence dashboard with sub-second querying and dynamic charting.

02 Architecture & Technical Implementation

Frontend: Developed a modular React.js dashboard utilizing custom hooks for state management and SVG/Canvas data visualization.

Backend: Designed lightweight Python data processing microservices to aggregate, filter, and calculate rolling window summary metrics.

Database: Engineered normalized MySQL schemas with indexing on timestamp and categorical dimensions to ensure instant telemetry response.

03 User Experience & Data Visualization Design

Created clean, high-density analytics cards with toggleable time intervals (24h, 7d, 30d, 1y).

Designed accessible color palettes with clear distinction between revenue growth, cost overruns, and projection baselines.

Implemented smooth keyframe transitions when drilling down into granular cohort breakdowns.

04 Outcomes & Performance Benchmarks

Dashboard load times reduced by 60% compared to legacy reporting portal.

Supported concurrent queries across 500,000+ transaction rows with median response latency under 120ms.

Adopted by 4 regional business units for daily revenue forecasting and executive standups.